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Chapter 27 - Power Series Methods

Constant-coefficient equations often reduce to algebra through a characteristic polynomial, a Laplace transform, or a matrix exponential. When coefficients depend on the independent variable, those routes may no longer produce a usable closed form.

Power series offer a different strategy. Instead of guessing the complete function, we build it coefficient by coefficient near a chosen centre.

This chapter will:

- review the series notation needed for derivatives and index shifts
- distinguish ordinary points from singular points
- derive recurrence formulas for homogeneous and forced equations
- separate the two arbitrary solution branches of a second-order equation
- construct series around points other than the origin
- apply initial conditions directly to series coefficients
- use repeated differentiation when only a few Taylor terms are needed

Turning An Unknown Function Into Unknown Coefficients

The Goal Of This Block

The goal is to understand what a power-series solution assumes and how differentiation changes its coefficients.

a power-series method replaces one unknown function by an ordered list of unknown numbers.

The Series Assumption

To seek a solution near $x = 0$, assume:

$$y(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Written term by term, this is:

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

Here:

- n is a nonnegative integer index
- a_n is the coefficient of x^n
- the numbers $a_0, a_1, a_2, \dots$ are initially unknown
- the series is centred at $x = 0$

The method succeeds when the differential equation forces these coefficients to satisfy a pattern called a recurrence relation.

Differentiate Without Skipping The First Terms

Start from the expanded series:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \cdots.$$

Differentiate term by term:

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \cdots.$$

Differentiate again:

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \cdots.$$

In summation notation:

$$y' = \sum_{n=1}^{\infty} na_nx^{n-1}$$

and:

$$y'' = \sum_{n=2}^{\infty} n(n-1)a_nx^{n-2}.$$

These forms are correct, but their powers do not yet begin with x^0 . That makes them awkward to compare with the original series.

Reindex The Derivatives To Powers Of X To The N

For y' , replace the old index by $n + 1$. Then:

$$y' = \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n.$$

Check the first terms:

$$n = 0 : \quad a_1,$$

$$n = 1 : \quad 2a_2x,$$

$$n = 2 : \quad 3a_3x^2.$$

For y'' , replace the old index by $n + 2$:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n.$$

Again, check the first terms:

$$n = 0 : \quad 2a_2,$$

$$n = 1 : \quad 6a_3x,$$

$$n = 2 : \quad 12a_4x^2.$$

The reindexed formulas reproduce the expanded derivatives exactly.

Why Matching Coefficients Is Valid

Suppose substitution produces:

$$c_0 + c_1x + c_2x^2 + c_3x^3 + \cdots = 0$$

throughout an interval around the centre. Then every coefficient must vanish:

$$c_0 = 0, \quad c_1 = 0, \quad c_2 = 0, \quad \dots$$

Equivalently, if:

$$\sum_{n=0}^{\infty} c_n x^n = 0,$$

then:

$$c_n = 0 \quad \text{for every } n \geq 0.$$

This is what converts a differential equation into algebraic equations for the coefficients a_n .¹

Initial Values Are The First Two Coefficients

From:

$$y(x) = a_0 + a_1 x + a_2 x^2 + \cdots,$$

set $x = 0$:

$$y(0) = a_0.$$

From:

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots,$$

set $x = 0$:

$$y'(0) = a_1.$$

¹ A power series has unique coefficients. In particular, differentiating n times and setting $x = 0$ gives $c_n = f^{(n)}(0)/n!$, so the zero function can have only zero coefficients.

Therefore:

$$\boxed{a_0 = y(0), \quad a_1 = y'(0)}.$$

For a second-order equation, these are the two free starting coefficients. The recurrence determines the remaining coefficients from them.

Block 1 Summary

Near the origin, use:

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n, \\ y' &= \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n, \\ y'' &= \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n. \end{aligned}$$

After every term is expressed with the same power, equality of power series allows coefficients to be matched.

Ordinary Points And Where The Method Applies

The Goal Of This Block

The goal is to decide whether an ordinary power series is justified at the chosen centre before doing recurrence algebra.

Analytic Functions

A function f is analytic at x_0 if its Taylor series about x_0 converges to the function throughout some interval around x_0 :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

Here $f^{(n)}$ is the n th derivative of f , and:

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

Polynomials, exponentials, sine, and cosine are analytic at every real point. A rational function is analytic wherever its denominator is nonzero. Being differentiable many times is related to analyticity but is not the definition. Analyticity requires the Taylor series to converge back to the function.

Normalize The Differential Equation First

Consider a second-order linear equation:

$$b_2(x)y'' + b_1(x)y' + b_0(x)y = g(x).$$

At points where $b_2(x) \neq 0$, divide every term by $b_2(x)$:

$$y'' + P(x)y' + Q(x)y = R(x),$$

where:

$$P(x) = \frac{b_1(x)}{b_2(x)}, \quad Q(x) = \frac{b_0(x)}{b_2(x)}, \quad R(x) = \frac{g(x)}{b_2(x)}.$$

The normalized coefficients P and Q , rather than the unnormalized appearance of the equation, determine the point type.

Ordinary And Singular Points

For the normalized homogeneous equation:

$$y'' + P(x)y' + Q(x)y = 0,$$

a point x_0 is an **ordinary point** when both P and Q are analytic at x_0 .

If either P or Q is not analytic at x_0 , then x_0 is a **singular point**.

For a forced equation, $R(x)$ should also be analytic at the centre if it is to be represented by an ordinary Taylor series there.

The practical meaning:

normalize first, inspect the coefficient functions at the intended centre, and only then assume an ordinary power series.

What An Ordinary Point Gives Us

If P and Q are analytic near x_0 , then each pair of starting values:

$$y(x_0) = a_0, \quad y'(x_0) = a_1$$

determines a unique local solution that is itself analytic near x_0 . For a forced equation, require R to be analytic there as well.

This matters because coefficient matching initially produces a formal list of numbers. The ordinary-point result tells us that the resulting power series converges to the actual local solution on some nonzero interval around the centre.²

An Ordinary-Point Check

Classify $x = 0$ for:

$$(1 + x^2)y'' + e^x y' + (x - 1)y = 0.$$

The leading coefficient at the centre is:

$$1 + 0^2 = 1 \neq 0,$$

so normalization is allowed near 0. Divide by $1 + x^2$:

$$y'' + \frac{e^x}{1 + x^2}y' + \frac{x - 1}{1 + x^2}y = 0.$$

² The guaranteed interval extends until analyticity can no longer be assured. Determining the exact radius may require examining singularities beyond the real line, which is not needed for the recurrence calculations in this chapter.

Thus:

$$P(x) = \frac{e^x}{1+x^2}, \quad Q(x) = \frac{x-1}{1+x^2}.$$

At $x = 0$, the common denominator is 1, not zero. Both ratios are analytic near 0. Therefore:

$x = 0 \text{ is an ordinary point.}$

A Singular-Point Check

Classify $x = 3$ for:

$$(x-3)y'' + (x+1)y' + 2y = 0.$$

For $x \neq 3$, divide by $x-3$:

$$y'' + \frac{x+1}{x-3}y' + \frac{2}{x-3}y = 0.$$

The normalized coefficients are:

$$P(x) = \frac{x+1}{x-3}, \quad Q(x) = \frac{2}{x-3}.$$

Both denominators vanish at $x = 3$. Therefore neither coefficient is analytic there, and:

$x = 3 \text{ is a singular point.}$

An ordinary Taylor-series assumption about $x = 3$ is not justified. A later method will treat an important class of singular points.

Do Not Confuse A Zero Coefficient With A Singular Point

In:

$$y'' + xy' + x^2y = 0,$$

the coefficients of y' and y happen to vanish at $x = 0$. That causes no problem. The equation is already normalized, with:

$$P(x) = x, \quad Q(x) = x^2.$$

Both are polynomials, so $x = 0$ is ordinary.

The dangerous zero is in the coefficient of the highest derivative before normalization, because dividing by that coefficient may create a pole.

Block 2 Summary

Normalize a second-order linear equation and inspect $P(x)$ and $Q(x)$. An ordinary point supports the ordinary power-series method; a singular point requires separate analysis.

Index Shifts And Coefficient Alignment

The Goal Of This Block

The goal is to make index shifting mechanical and checkable before using it in a complete differential equation.

Multiplication By X Shifts Powers Up

Start from:

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots.$$

Multiply by x :

$$xy = a_0 x + a_1 x^2 + a_2 x^3 + \cdots.$$

The result has no constant term. To write its coefficient of x^n , observe:

$$x^n \text{ receives } a_{n-1} \text{ for } n \geq 1.$$

Therefore:

$$xy = \sum_{n=1}^{\infty} a_{n-1} x^n.$$

The lower limit is 1, not 0, because a_{-1} has not been defined.

Multiplication By X Applied To A Derivative

Use:

$$y' = a_1 + 2a_2x + 3a_3x^2 + \cdots.$$

Multiply by x :

$$xy' = a_1x + 2a_2x^2 + 3a_3x^3 + \cdots.$$

For the coefficient of x^n , the pattern is na_n . Hence:

$$xy' = \sum_{n=1}^{\infty} na_nx^n.$$

There is again no constant term.

Multiplying Two Full Power Series

An analytic coefficient may have its own series:

$$P(x) = p_0 + p_1x + p_2x^2 + \cdots = \sum_{n=0}^{\infty} p_nx^n.$$

Multiply it by:

$$y(x) = a_0 + a_1x + a_2x^2 + \cdots.$$

Collect the first three powers explicitly:

$$\begin{aligned} P(x)y(x) &= p_0a_0 \\ &\quad + (p_0a_1 + p_1a_0)x \\ &\quad + (p_0a_2 + p_1a_1 + p_2a_0)x^2 + \cdots. \end{aligned}$$

For the coefficient of x^n , pair every $p_k x^k$ with the term $a_{n-k} x^{n-k}$. Their powers add to n . Therefore:

$$P(x)y(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n p_k a_{n-k} \right) x^n.$$

The inner finite sum is the coefficient convolution. When $P(x) = x$, only $p_1 = 1$ is nonzero, and the formula reduces to the earlier shift a_{n-1} for $n \geq 1$.

Separate Low Powers When Lower Limits Differ

Suppose an equation produces:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} a_{n-1}x^n = 0.$$

The first sum contains an x^0 term, but the second does not. Extract the constant term from the first sum:

$$2a_2 + \sum_{n=1}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} a_{n-1}x^n = 0.$$

Now combine the sums that share the lower limit $n = 1$:

$$2a_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + a_{n-1}] x^n = 0.$$

Match the constant term separately:

$$2a_2 = 0.$$

For every $n \geq 1$, match the coefficient of x^n :

$$(n+2)(n+1)a_{n+2} + a_{n-1} = 0.$$

This separation prevents an undefined a_{-1} from entering the recurrence.

A Quick Index Check

For any reindexed sum, test the first two terms.

For example:

$$\sum_{n=1}^{\infty} a_{n-1} x^n$$

starts with:

$$n = 1 : a_0 x, \quad n = 2 : a_1 x^2.$$

These are exactly the first two terms of xy . If the expanded terms do not match, the index shift is wrong.

Block 3 Summary

Do not combine sums until their powers and lower limits agree. Expand a few terms, reindex, test the first terms, and separate exceptional low powers before deriving a recurrence.

A Complete Homogeneous Series Solution

The Goal Of This Block

The goal is to derive and use a recurrence formula for a variable-coefficient homogeneous equation without skipping coefficient substitutions.

The Problem And Point Check

Find a power-series solution near $x = 0$ for:

$$y'' + xy' + 2y = 0.$$

The equation is already normalized:

$$P(x) = x, \quad Q(x) = 2.$$

Both are analytic at 0, so $x = 0$ is an ordinary point.

Substitute All Three Series

Use the three required series:

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^n, \\ y'' &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n, \\ xy' &= \sum_{n=1}^{\infty} na_n x^n. \end{aligned}$$

Substitute into $y'' + xy' + 2y = 0$:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} na_nx^n + 2 \sum_{n=0}^{\infty} a_nx^n = 0.$$

Although the xy' sum begins at $n = 1$, its missing $n = 0$ term would be $0a_0x^0 = 0$. We may therefore include that zero term and write:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + na_n + 2a_n]x^n = 0.$$

Combine the two coefficients multiplying a_n :

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+2)a_n]x^n = 0.$$

Derive The Recurrence

Every coefficient of x^n must equal zero:

$$(n+2)(n+1)a_{n+2} + (n+2)a_n = 0, \quad n \geq 0.$$

Move the a_n term to the right:

$$(n+2)(n+1)a_{n+2} = -(n+2)a_n.$$

Divide by $(n+2)(n+1)$:

$$a_{n+2} = -\frac{n+2}{(n+2)(n+1)}a_n.$$

Cancel the nonzero factor $n + 2$:

$$a_{n+2} = -\frac{a_n}{n+1}, \quad n \geq 0.$$

The domain $n \geq 0$ is part of the recurrence formula.

Build The Even Coefficients

Set $n = 0$ in the recurrence:

$$\begin{aligned} a_2 &= -\frac{a_0}{0+1} \\ &= -a_0. \end{aligned}$$

Set $n = 2$:

$$\begin{aligned} a_4 &= -\frac{a_2}{2+1} \\ &= -\frac{-a_0}{3} \\ &= \frac{a_0}{3}. \end{aligned}$$

Continue the even chain by using $a_{n+2} = -a_n/(n+1)$ with $n = 4$:

$$\begin{aligned} a_6 &= -\frac{a_4}{4+1} \\ &= -\frac{a_0/3}{5} \\ &= -\frac{a_0}{15}. \end{aligned}$$

Thus the even branch begins:

$$a_0 \left(1 - x^2 + \frac{x^4}{3} - \frac{x^6}{15} + \cdots \right).$$

Build The Odd Coefficients

Set $n = 1$:

$$\begin{aligned} a_3 &= -\frac{a_1}{1+1} \\ &= -\frac{a_1}{2}. \end{aligned}$$

Set $n = 3$:

$$\begin{aligned} a_5 &= -\frac{a_3}{3+1} \\ &= -\frac{-a_1/2}{4} \\ &= \frac{a_1}{8}. \end{aligned}$$

Set $n = 5$:

$$\begin{aligned} a_7 &= -\frac{a_5}{5+1} \\ &= -\frac{a_1/8}{6} \\ &= -\frac{a_1}{48}. \end{aligned}$$

Thus the odd branch begins:

$$a_1 \left(x - \frac{x^3}{2} + \frac{x^5}{8} - \frac{x^7}{48} + \cdots \right).$$

Assemble The General Solution

Return to:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots.$$

Insert the calculated even and odd coefficients:

$$y(x) = a_0 \left(1 - x^2 + \frac{x^4}{3} - \frac{x^6}{15} + \cdots \right) + a_1 \left(x - \frac{x^3}{2} + \frac{x^5}{8} - \frac{x^7}{48} + \cdots \right).$$

The constants a_0 and a_1 remain arbitrary. They are precisely $y(0)$ and $y'(0)$.

Why Two Series Appear

The recurrence advances by two indices:

$$a_n \longrightarrow a_{n+2}.$$

Therefore even coefficients depend only on a_0 , while odd coefficients depend only on a_1 . The two chains never mix.

This is how a second-order equation preserves two independent arbitrary constants inside the series method.

Block 4 Summary

For $y'' + xy' + 2y = 0$, coefficient matching gives:

$$a_{n+2} = -\frac{a_n}{n+1}, \quad n \geq 0.$$

Applying the recurrence separately to even and odd indices produces two solution branches controlled by a_0 and a_1 .

Nonhomogeneous Equations And Exceptional Powers

The Goal Of This Block

The goal is to match a series on the left to a nonzero series on the right and handle low powers that do not follow the later recurrence.

The Forced Problem

Find the first several terms of the general power-series solution near $x = 0$ for:

$$y'' + xy = 1 + x.$$

The equation is normalized, with $P(x) = 0$ and $Q(x) = x$. Both are analytic at 0. The forcing $1 + x$ is also analytic there.

Substitute And Align The Powers

Use:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n.$$

For xy :

$$xy = \sum_{n=1}^{\infty} a_{n-1}x^n.$$

Substitute both series:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} a_{n-1}x^n = 1 + x.$$

Extract the constant term from the first sum:

$$2a_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + a_{n-1}]x^n = 1 + x.$$

The right side has coefficients:

$$1 + x = 1 \cdot x^0 + 1 \cdot x^1 + 0 \cdot x^2 + 0 \cdot x^3 + \dots.$$

Match The Constant And Linear Terms First

Match the coefficient of x^0 :

$$2a_2 = 1.$$

Divide by 2:

$$a_2 = \frac{1}{2}.$$

Match the coefficient of x^1 . Set $n = 1$ in the combined sum:

$$(1+2)(1+1)a_{1+2} + a_{1-1} = 1.$$

Simplify the indices and factors:

$$6a_3 + a_0 = 1.$$

Subtract a_0 and divide by 6:

$$a_3 = \frac{1 - a_0}{6}.$$

These equations are special because the forcing has nonzero constant and linear coefficients.

Derive The Later Recurrence

For every $n \geq 2$, the coefficient of x^n on the right is zero. Therefore:

$$(n+2)(n+1)a_{n+2} + a_{n-1} = 0, \quad n \geq 2.$$

Move a_{n-1} to the right:

$$(n+2)(n+1)a_{n+2} = -a_{n-1}.$$

Divide by $(n+2)(n+1)$:

$$a_{n+2} = -\frac{a_{n-1}}{(n+2)(n+1)}, \quad n \geq 2.$$

This recurrence is not valid for $n = 0$ or $n = 1$. Those two powers were matched separately.

Calculate The Next Coefficients

Set $n = 2$:

$$\begin{aligned} a_4 &= -\frac{a_1}{(4)(3)} \\ &= -\frac{a_1}{12}. \end{aligned}$$

Set $n = 3$:

$$\begin{aligned} a_5 &= -\frac{a_2}{(5)(4)} \\ &= -\frac{1/2}{20} \\ &= -\frac{1}{40}. \end{aligned}$$

Set $n = 4$:

$$\begin{aligned} a_6 &= -\frac{a_3}{(6)(5)} \\ &= -\frac{(1 - a_0)/6}{30} \\ &= -\frac{1 - a_0}{180}. \end{aligned}$$

Assemble And Interpret The Series

Insert the calculated coefficients into:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots.$$

This gives:

$$\begin{aligned} y(x) &= a_0 + a_1x + \frac{1}{2}x^2 + \frac{1 - a_0}{6}x^3 - \frac{a_1}{12}x^4 \\ &\quad - \frac{1}{40}x^5 - \frac{1 - a_0}{180}x^6 + \dots. \end{aligned}$$

Group the terms multiplying a_0 , the terms multiplying a_1 , and the terms independent of both:

$$\begin{aligned} y(x) = & a_0 \left(1 - \frac{x^3}{6} + \frac{x^6}{180} + \cdots \right) \\ & + a_1 \left(x - \frac{x^4}{12} + \cdots \right) \\ & + \left(\frac{x^2}{2} + \frac{x^3}{6} - \frac{x^5}{40} - \frac{x^6}{180} + \cdots \right). \end{aligned}$$

The first two lines form the general homogeneous response. The third line is one particular response to $1 + x$.

Block 5 Summary

For a nonhomogeneous equation, expand the right side as a series and match its coefficients. If its first few coefficients differ from the later pattern, solve those powers separately and state where the general recurrence begins.

Series About A Nonzero Centre

The Goal Of This Block

The goal is to centre a series at $x = x_0$ and translate the algebra to the origin without changing the derivatives incorrectly.

The Shifted Series

A power series centred at x_0 has the form:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

Its first two coefficients satisfy:

$$a_0 = y(x_0), \quad a_1 = y'(x_0).$$

Working directly with repeated powers of $(x - x_0)$ is possible, but a change of variable makes the notation cleaner.

Translate The Centre To Zero

Define:

$$t = x - x_0.$$

Then:

$$x = t + x_0.$$

Since $dt/dx = 1$, the chain rule gives:

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt}.$$

Differentiate once more:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dt} \right) = \frac{d^2y}{dt^2} \frac{dt}{dx} = \frac{d^2y}{dt^2}.$$

Thus the derivative symbols keep the same numerical values under a pure translation. The coefficient functions must still have $x = t + x_0$ substituted into them.

A Complete Shifted-Centre Example

Find a series solution near $x = 2$ for:

$$y'' + (x - 2)y' - y = 0.$$

Set:

$$t = x - 2.$$

Then $x - 2 = t$, and the equation becomes:

$$\frac{d^2y}{dt^2} + t \frac{dy}{dt} - y = 0.$$

To keep the notation short, write derivatives with respect to t as primes:

$$y'' + ty' - y = 0.$$

The transformed equation is to be solved near $t = 0$.

Derive The Shifted Recurrence

Assume:

$$y = \sum_{n=0}^{\infty} a_n t^n.$$

Then:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}t^n$$

and:

$$ty' = \sum_{n=1}^{\infty} na_n t^n.$$

As before, the missing $n = 0$ term in ty' is zero. Substitute all terms:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + na_n - a_n] t^n = 0.$$

Combine the last two terms:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n-1)a_n] t^n = 0.$$

Set each coefficient equal to zero:

$$(n+2)(n+1)a_{n+2} + (n-1)a_n = 0.$$

Solve for a_{n+2} :

$$a_{n+2} = \frac{1-n}{(n+2)(n+1)} a_n, \quad n \geq 0.$$

Generate And Return To X

For the even chain, set $n = 0$:

$$a_2 = \frac{1}{2} a_0.$$

Set $n = 2$:

$$\begin{aligned} a_4 &= \frac{1-2}{(4)(3)} a_2 \\ &= -\frac{1}{12} \left(\frac{a_0}{2} \right) \\ &= -\frac{a_0}{24}. \end{aligned}$$

For the odd chain, use the same recurrence with $n = 1$:

$$a_3 = \frac{1-1}{(3)(2)} a_1 = 0.$$

Once $a_3 = 0$, the recurrence gives $a_5 = a_7 = \dots = 0$. Therefore:

$$y(t) = a_0 \left(1 + \frac{t^2}{2} - \frac{t^4}{24} + \dots \right) + a_1 t.$$

Return to $t = x - 2$:

$$y(x) = a_0 \left[1 + \frac{(x-2)^2}{2} - \frac{(x-2)^4}{24} + \dots \right] + a_1(x-2).$$

The powers must remain powers of $(x - 2)$ because the series is centred at 2.

Apply Initial Conditions At The Centre

Suppose the shifted-centre example also has:

$$y(2) = 3, \quad y'(2) = -2.$$

Because:

$$a_0 = y(2), \quad a_1 = y'(2),$$

the conditions give immediately:

$$a_0 = 3, \quad a_1 = -2.$$

Substitute these values into the general series:

$$y(x) = 3 + \frac{3}{2}(x-2)^2 - \frac{1}{8}(x-2)^4 - 2(x-2) + \dots.$$

Reordering by increasing power gives:

$$y(x) = 3 - 2(x-2) + \frac{3}{2}(x-2)^2 - \frac{1}{8}(x-2)^4 + \dots.$$

Block 6 Summary

For a centre x_0 , set $t = x - x_0$, substitute $x = t + x_0$ throughout the equation, solve near $t = 0$, and finally replace t by $x - x_0$. Initial values given at the centre become a_0 and a_1 directly.

A Direct Taylor Method For Initial-Value Problems

The Goal Of This Block

The goal is to calculate a few terms of an IVP solution by finding successive derivatives at the initial point.

The Taylor Formula At The Initial Point

If the solution is analytic near x_0 , then:

$$y(x) = y(x_0) + y'(x_0)(x - x_0) + \frac{y''(x_0)}{2!}(x - x_0)^2 + \frac{y'''(x_0)}{3!}(x - x_0)^3 + \dots$$

The initial conditions supply $y(x_0)$ and $y'(x_0)$. The differential equation supplies $y''(x_0)$, and differentiated versions of the equation supply later derivatives.

A Four-Term IVP Calculation

Find the Taylor polynomial through degree four for the solution of:

$$y'' = xy' - 2y + x^2, \quad y(1) = 2, \quad y'(1) = -1.$$

The centre is $x_0 = 1$. We already know:

$$y(1) = 2, \quad y'(1) = -1.$$

Find The Second Derivative

Use the differential equation itself:

$$y'' = xy' - 2y + x^2.$$

Set $x = 1$, $y(1) = 2$, and $y'(1) = -1$:

$$\begin{aligned} y''(1) &= (1)(-1) - 2(2) + 1^2 \\ &= -1 - 4 + 1 \\ &= -4. \end{aligned}$$

Find The Third Derivative

Differentiate every term in:

$$y'' = xy' - 2y + x^2.$$

The product rule gives $(xy')' = y' + xy''$. Therefore:

$$\begin{aligned} y''' &= y' + xy'' - 2y' + 2x \\ &= xy'' - y' + 2x. \end{aligned}$$

Set $x = 1$, $y''(1) = -4$, and $y'(1) = -1$:

$$\begin{aligned} y'''(1) &= (1)(-4) - (-1) + 2(1) \\ &= -4 + 1 + 2 \\ &= -1. \end{aligned}$$

Find The Fourth Derivative

Differentiate the simplified third-derivative equation:

$$y''' = xy'' - y' + 2x.$$

Apply the product rule to xy'' :

$$(xy'')' = y'' + xy'''.$$

Therefore:

$$\begin{aligned} y'''' &= y'' + xy''' - y'' + 2 \\ &= xy''' + 2. \end{aligned}$$

Set $x = 1$ and $y'''(1) = -1$:

$$\begin{aligned} y''''(1) &= (1)(-1) + 2 \\ &= 1. \end{aligned}$$

Build The Taylor Polynomial

Use:

$$\begin{aligned} y(1) &= 2, \\ y'(1) &= -1, \\ y''(1) &= -4, \\ y'''(1) &= -1, \\ y''''(1) &= 1. \end{aligned}$$

Substitute these values into the Taylor formula centred at 1:

$$\begin{aligned} y(x) &= 2 - (x - 1) + \frac{-4}{2!}(x - 1)^2 + \frac{-1}{3!}(x - 1)^3 + \frac{1}{4!}(x - 1)^4 + \dots \\ &= 2 - (x - 1) - 2(x - 1)^2 - \frac{1}{6}(x - 1)^3 + \frac{1}{24}(x - 1)^4 + \dots \end{aligned}$$

Thus the requested degree-four Taylor polynomial is:

$$T_4(x) = 2 - (x - 1) - 2(x - 1)^2 - \frac{1}{6}(x - 1)^3 + \frac{1}{24}(x - 1)^4.$$

Recurrence Method Or Direct Taylor Method?

Use coefficient matching when:

- a general recurrence is needed
- many terms are required
- the two arbitrary solution branches matter
- the structure of the entire series is the goal

Use repeated differentiation when:

- initial values are already given
- only the first few terms are required
- evaluating derivatives at the centre stays manageable

The direct Taylor method is efficient locally, but it may not reveal a clean formula for the general coefficient a_n .

Block 7 Summary

For an IVP, successive differentiation of the differential equation can generate $y''(x_0), y'''(x_0), \dots$. Insert those values into the Taylor formula and keep the factorial denominators visible.

A Reliable Workflow And Common Mistakes

The Goal Of This Block

The goal is to turn power-series work into a repeatable sequence and identify the index errors that most often corrupt a recurrence.

The Power-Series Workflow

Step 1: Choose and check the centre. Normalize the equation and confirm that the centre is ordinary. For a forced equation, check the forcing as well.

Step 2: Translate a nonzero centre. If the centre is $x_0 \neq 0$, use $t = x - x_0$ and replace every occurrence of x by $t + x_0$.

Step 3: Write the series and derivatives. State $y, y',$ and y'' with their indices and lower limits.

Step 4: Substitute before combining. Keep one sum for each differential-equation term until every multiplication by $x, x^2,$ or another series has been handled.

Step 5: Align powers and lower limits. Reindex to a common power. Extract low-order terms when a sum starts later than the others.

Step 6: Match coefficients. For a homogeneous equation, match each coefficient to zero. For a forced equation, match it to the corresponding coefficient on the right.

Step 7: Solve for the highest-index coefficient. State both the recurrence and the values of n for which it is valid.

Step 8: Generate enough coefficients. Show each substitution until the pattern is unmistakable. Keep even and odd chains separate when the recurrence advances by two.

Step 9: Apply conditions and reconstruct. Insert the coefficients into the original series and return from any shifted variable to x .

Step 10: Check the requested order. If terms through x^m are requested, calculate every coefficient through a_m , including coefficients that turn out to be zero.

Common Mistake: Matching Different Powers

The sums:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} \quad \text{and} \quad \sum_{n=0}^{\infty} a_n x^n$$

cannot be combined as written because one uses x^{n-2} and the other uses x^n . Reindex the first sum:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n.$$

Only then do corresponding coefficients refer to the same power.

Common Mistake: Ignoring The Lower Limit

For:

$$xy = \sum_{n=1}^{\infty} a_{n-1} x^n,$$

changing the lower limit to 0 would introduce a_{-1} , which has not been defined.

Either preserve the lower limit 1 and separate the constant term, or explicitly define a temporary zero coefficient. The first route is usually clearer for learners.

Common Mistake: Hiding The Recurrence Domain

A formula involving a_{n-1} may be valid only for $n \geq 1$ or $n \geq 2$. Applying it at $n = 0$ can create an undefined coefficient or miss a nonzero forcing term.

Write the range beside the recurrence:

$$a_{n+2} = \dots, \quad n \geq 2.$$

Then solve the exceptional powers separately.

Common Mistake: Losing A Free Constant

A second-order homogeneous equation should normally retain two free coefficients. If a recurrence advancing by two is calculated only from a_0 , the odd branch beginning with a_1 has been omitted.

Write both chains:

$$a_0 \rightarrow a_2 \rightarrow a_4 \rightarrow \dots, \quad a_1 \rightarrow a_3 \rightarrow a_5 \rightarrow \dots.$$

Common Mistake: Returning To The Wrong Variable

If $t = x - x_0$, then a term $a_3 t^3$ becomes:

$$a_3(x - x_0)^3,$$

not $a_3 x^3$. Preserve the shifted power when returning to the original variable.

Block 8 Summary

Power-series algebra is reliable when powers, lower limits, recurrence ranges, and free coefficients remain visible. Most errors come from combining sums too early or using a recurrence outside its stated domain.

Original Practice Set

Problems 1 To 4: Foundations And A First Solution

Problem 1: Reindex Two Derivatives. Starting from:

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

derive formulas for y' and y'' in powers of x^n . Check the first two terms of each result.

Problem 2: Classify Two Points. For:

$$(4 - x)y'' + (1 + x)y' + e^x y = 0,$$

classify $x = 0$ and $x = 4$ as ordinary or singular points.

Problem 3: Derive And Use A Four-Step Recurrence. Find the recurrence and the first three nonzero terms in each solution branch near $x = 0$ for:

$$y'' + x^2y = 0.$$

Problem 4: Apply Initial Values To A Known Recurrence. Use the recurrence:

$$a_{n+2} = -\frac{a_n}{n+1}, \quad n \geq 0,$$

to find the series through x^5 satisfying:

$$y'' + xy' + 2y = 0, \quad y(0) = 2, \quad y'(0) = -1.$$

Problems 5 To 8: Forcing And Shifted Centres

Problem 5: Separate A Missing Low-Power Term. Find a recurrence and write the general solution through x^6 for:

$$y'' - xy = 0$$

near $x = 0$.

Problem 6: Match A Polynomial Forcing. For:

$$y'' + y = x^2,$$

find a_2, a_3, a_4 , and a_5 in terms of the arbitrary coefficients a_0 and a_1 .

Problem 7: Translate To A New Centre. Set $t = x + 1$ and find the series through degree six near $x = -1$ for:

$$y'' + (x + 1)y = 0.$$

Leave the two arbitrary coefficients visible.

Problem 8: Use Conditions At A Shifted Centre. Find the series through degree four near $x = 1$ for:

$$y'' + (x - 1)y' - y = 0, \quad y(1) = 2, \quad y'(1) = 3.$$

Problems 9 To 12: Taylor Data And Error Diagnosis

Problem 9: Build A Taylor Polynomial Directly. Use repeated differentiation to find the degree-four Taylor polynomial about $x = 0$ for:

$$y'' = y' + xy, \quad y(0) = 1, \quad y'(0) = 2.$$

Problem 10: Confirm The Same Polynomial By Recurrence. For the IVP in Problem 9, derive a coefficient recurrence and verify the coefficients through a_4 .

Problem 11: Diagnose Four Series Errors. Explain the error in each statement.

1. Since $y'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$, its coefficient of x^n is $n(n-1)a_n$.
2. The identity $xy = \sum_{n=0}^{\infty} a_{n-1}x^n$ is valid without any additional definition.
3. A recurrence containing a_{n-1} can always be evaluated at $n = 0$.
4. A second-order homogeneous series containing only a_0 is automatically the general solution.

Problem 12: Evaluate A Truncated Series. Use the degree-five series from Problem 4 to approximate $y(0.2)$. Keep at least six decimal places during the calculation.

Worked Answers: Problems 1 To 4**Answer 1.** Differentiate:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Replace the old index n by $n + 1$ so the power becomes x^n :

$$y' = \sum_{n=0}^{\infty} (n + 1) a_{n+1} x^n.$$

Its first two terms are:

$$n = 0 : a_1, \quad n = 1 : 2a_2x.$$

Differentiate a second time:

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Replace the old index by $n + 2$:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

Its first two terms are:

$$n = 0 : 2a_2, \quad n = 1 : 6a_3x.$$

These agree with direct differentiation of $a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$.

Answer 2. Divide by the leading coefficient $4 - x$:

$$y'' + \frac{1+x}{4-x}y' + \frac{e^x}{4-x}y = 0.$$

Thus:

$$P(x) = \frac{1+x}{4-x}, \quad Q(x) = \frac{e^x}{4-x}.$$

At $x = 0$, the denominator is:

$$4 - 0 = 4 \neq 0.$$

Both normalized coefficients are analytic near 0, so:

$x = 0 \text{ is an ordinary point.}$

At $x = 4$, the denominator is zero. Both normalized coefficients fail to be analytic there, so:

$x = 4 \text{ is a singular point.}$

Answer 3. Use:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n$$

and:

$$x^2y = \sum_{n=2}^{\infty} a_{n-2}x^n.$$

Substitute into $y'' + x^2y = 0$:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=2}^{\infty} a_{n-2}x^n = 0.$$

The second sum has no constant or linear term. Match those powers from the first sum:

$$2a_2 = 0, \quad 6a_3 = 0.$$

Therefore:

$$a_2 = 0, \quad a_3 = 0.$$

For $n \geq 2$:

$$(n+2)(n+1)a_{n+2} + a_{n-2} = 0.$$

Solve for the highest-index coefficient:

$$a_{n+2} = -\frac{a_{n-2}}{(n+2)(n+1)}, \quad n \geq 2.$$

Set $n = 2$ and then $n = 6$ for the branch beginning with a_0 :

$$a_4 = -\frac{a_0}{(4)(3)} = -\frac{a_0}{12},$$

$$a_8 = -\frac{a_4}{(8)(7)} = \frac{a_0}{672}.$$

Set $n = 3$ and then $n = 7$ for the branch beginning with a_1 :

$$a_5 = -\frac{a_1}{(5)(4)} = -\frac{a_1}{20},$$

$$a_9 = -\frac{a_5}{(9)(8)} = \frac{a_1}{1440}.$$

Hence:

$$y(x) = a_0 \left(1 - \frac{x^4}{12} + \frac{x^8}{672} + \cdots \right) + a_1 \left(x - \frac{x^5}{20} + \frac{x^9}{1440} + \cdots \right).$$

Answer 4. The initial conditions give:

$$a_0 = y(0) = 2, \quad a_1 = y'(0) = -1.$$

Use the recurrence with $n = 0$:

$$a_2 = -\frac{a_0}{1}$$

$$= -2.$$

With $n = 1$:

$$a_3 = -\frac{a_1}{2}$$

$$= \frac{1}{2}.$$

With $n = 2$:

$$\begin{aligned}a_4 &= -\frac{a_2}{3} \\&= \frac{2}{3}.\end{aligned}$$

With $n = 3$:

$$\begin{aligned}a_5 &= -\frac{a_3}{4} \\&= -\frac{1}{8}.\end{aligned}$$

Insert the coefficients into the series:

$$y(x) = 2 - x - 2x^2 + \frac{1}{2}x^3 + \frac{2}{3}x^4 - \frac{1}{8}x^5 + \cdots.$$

Worked Answers: Problems 5 To 8

Answer 5. Use:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n$$

and:

$$xy = \sum_{n=1}^{\infty} a_{n-1}x^n$$

Substitute both series into $y'' - xy = 0$:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_{n=1}^{\infty} a_{n-1}x^n = 0.$$

The constant term comes only from y'' :

$$2a_2 = 0, \quad a_2 = 0.$$

For $n \geq 1$:

$$(n+2)(n+1)a_{n+2} - a_{n-1} = 0.$$

Therefore:

$$a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)}, \quad n \geq 1.$$

Set $n = 1$:

$$a_3 = \frac{a_0}{(3)(2)} = \frac{a_0}{6}.$$

Set $n = 2$:

$$a_4 = \frac{a_1}{(4)(3)} = \frac{a_1}{12}.$$

Set $n = 3$:

$$a_5 = \frac{a_2}{(5)(4)} = 0.$$

Continue the recurrence $a_{n+2} = a_{n-1}/[(n+2)(n+1)]$ with $n = 4$:

$$\begin{aligned} a_6 &= \frac{a_3}{(6)(5)} \\ &= \frac{a_0/6}{30} \\ &= \frac{a_0}{180}. \end{aligned}$$

Thus, through degree six:

$$y(x) = a_0 \left(1 + \frac{x^3}{6} + \frac{x^6}{180} \right) + a_1 \left(x + \frac{x^4}{12} \right) + \cdots.$$

Answer 6. Write the forcing as a power series with coefficients:

$$x^2 = 0 + 0x + 1x^2 + 0x^3 + \cdots.$$

Substitution into $y'' + y = x^2$ gives:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n] x^n = x^2.$$

Match the constant coefficient:

$$2a_2 + a_0 = 0,$$

so:

$$a_2 = -\frac{a_0}{2}.$$

Match the coefficient of x^1 :

$$6a_3 + a_1 = 0,$$

so:

$$a_3 = -\frac{a_1}{6}.$$

Match the coefficient of x^2 , which is 1 on the right:

$$12a_4 + a_2 = 1.$$

Substitute $a_2 = -a_0/2$:

$$\begin{aligned} 12a_4 - \frac{a_0}{2} &= 1, \\ 12a_4 &= 1 + \frac{a_0}{2}, \\ a_4 &= \frac{1}{12} + \frac{a_0}{24}. \end{aligned}$$

Match the coefficient of x^3 , which is zero on the right:

$$20a_5 + a_3 = 0.$$

Substitute $a_3 = -a_1/6$:

$$\begin{aligned} 20a_5 - \frac{a_1}{6} &= 0, \\ a_5 &= \frac{a_1}{120}. \end{aligned}$$

Therefore:

$$a_2 = -\frac{a_0}{2}, \quad a_3 = -\frac{a_1}{6}, \quad a_4 = \frac{1}{12} + \frac{a_0}{24}, \quad a_5 = \frac{a_1}{120}.$$

Answer 7. Set:

$$t = x + 1.$$

Then $x + 1 = t$, and translations preserve the first and second derivatives. The equation becomes:

$$y'' + ty = 0$$

near $t = 0$.

Substitute:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}t^n$$

and:

$$ty = \sum_{n=1}^{\infty} a_{n-1}t^n.$$

The constant term gives:

$$2a_2 = 0, \quad a_2 = 0.$$

For $n \geq 1$:

$$(n+2)(n+1)a_{n+2} + a_{n-1} = 0,$$

so:

$$a_{n+2} = -\frac{a_{n-1}}{(n+2)(n+1)}.$$

Calculate the needed coefficients:

$$a_3 = -\frac{a_0}{6},$$

$$a_4 = -\frac{a_1}{12},$$

$$a_5 = -\frac{a_2}{20} = 0,$$

$$a_6 = -\frac{a_3}{30} = \frac{a_0}{180}.$$

Thus:

$$y(t) = a_0 \left(1 - \frac{t^3}{6} + \frac{t^6}{180} \right) + a_1 \left(t - \frac{t^4}{12} \right) + \dots$$

Return to $t = x + 1$:

$$\boxed{\begin{aligned} y(x) = a_0 & \left[1 - \frac{(x+1)^3}{6} + \frac{(x+1)^6}{180} \right] \\ & + a_1 \left[(x+1) - \frac{(x+1)^4}{12} \right] + \dots \end{aligned}}$$

Answer 8. Set:

$$t = x - 1.$$

The equation becomes:

$$y'' + ty' - y = 0.$$

Its recurrence is:

$$a_{n+2} = \frac{1-n}{(n+2)(n+1)}a_n, \quad n \geq 0.$$

The initial conditions at the centre give:

$$a_0 = y(1) = 2, \quad a_1 = y'(1) = 3.$$

Set $n = 0$:

$$\begin{aligned} a_2 &= \frac{1}{(2)(1)}a_0 \\ &= 1. \end{aligned}$$

Set $n = 1$:

$$a_3 = \frac{0}{(3)(2)}a_1 = 0.$$

Set $n = 2$:

$$\begin{aligned} a_4 &= \frac{-1}{(4)(3)}a_2 \\ &= -\frac{1}{12}. \end{aligned}$$

With $a_0 = 2$, $a_1 = 3$, $a_2 = 1$, $a_3 = 0$, and $a_4 = -1/12$, the degree-four series in t is:

$$y(t) = 2 + 3t + t^2 - \frac{1}{12}t^4 + \dots$$

Replace t by $x - 1$:

$$y(x) = 2 + 3(x - 1) + (x - 1)^2 - \frac{1}{12}(x - 1)^4 + \dots$$

Worked Answers: Problems 9 To 12

Answer 9. The initial values are:

$$y(0) = 1, \quad y'(0) = 2.$$

Use the differential equation:

$$y'' = y' + xy.$$

Set $x = 0$:

$$\begin{aligned} y''(0) &= y'(0) + 0 \cdot y(0) \\ &= 2. \end{aligned}$$

Differentiate the differential equation:

$$\begin{aligned} y''' &= y'' + (xy)' \\ &= y'' + y + xy'. \end{aligned}$$

Set $x = 0$:

$$\begin{aligned}y'''(0) &= y''(0) + y(0) + 0 \cdot y'(0) \\&= 2 + 1 \\&= 3.\end{aligned}$$

Differentiate again:

$$\begin{aligned}y'''' &= y''' + y' + (xy')' \\&= y''' + y' + y' + xy'' \\&= y''' + 2y' + xy''.\end{aligned}$$

Set $x = 0$:

$$\begin{aligned}y''''(0) &= y'''(0) + 2y'(0) + 0 \cdot y''(0) \\&= 3 + 2(2) \\&= 7.\end{aligned}$$

Insert the derivatives into the Taylor formula:

$$\begin{aligned}T_4(x) &= 1 + 2x + \frac{2}{2!}x^2 + \frac{3}{3!}x^3 + \frac{7}{4!}x^4 \\&= 1 + 2x + x^2 + \frac{1}{2}x^3 + \frac{7}{24}x^4.\end{aligned}$$

Thus:

$$T_4(x) = 1 + 2x + x^2 + \frac{1}{2}x^3 + \frac{7}{24}x^4.$$

Answer 10. Rewrite the equation as:

$$y'' - y' - xy = 0.$$

Use:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n,$$

$$y' = \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n,$$

$$xy = \sum_{n=1}^{\infty} a_{n-1}x^n.$$

The constant term gives:

$$2a_2 - a_1 = 0.$$

Therefore:

$$a_2 = \frac{a_1}{2}.$$

For $n \geq 1$:

$$(n+2)(n+1)a_{n+2} - (n+1)a_{n+1} - a_{n-1} = 0.$$

Solve for a_{n+2} :

$$a_{n+2} = \frac{(n+1)a_{n+1} + a_{n-1}}{(n+2)(n+1)}, \quad n \geq 1.$$

The initial values give:

$$a_0 = 1, \quad a_1 = 2.$$

Then:

$$a_2 = \frac{2}{2} = 1.$$

Set $n = 1$:

$$\begin{aligned} a_3 &= \frac{2a_2 + a_0}{(3)(2)} \\ &= \frac{2(1) + 1}{6} \\ &= \frac{1}{2}. \end{aligned}$$

Set $n = 2$:

$$\begin{aligned} a_4 &= \frac{3a_3 + a_1}{(4)(3)} \\ &= \frac{3(1/2) + 2}{12} \\ &= \frac{7/2}{12} \\ &= \frac{7}{24}. \end{aligned}$$

Thus the recurrence gives:

$$y(x) = 1 + 2x + x^2 + \frac{1}{2}x^3 + \frac{7}{24}x^4 + \cdots,$$

which agrees with the direct Taylor calculation.

Answer 11.

Statement 1. The expression $n(n-1)a_n$ multiplies x^{n-2} , not x^n . Reindexing gives the coefficient of x^n as:

$$(n+2)(n+1)a_{n+2}.$$

Statement 2. At $n = 0$, the proposed sum contains a_{-1} , which has not been defined. The standard form is:

$$xy = \sum_{n=1}^{\infty} a_{n-1}x^n.$$

Statement 3. Substituting $n = 0$ into a recurrence containing a_{n-1} produces a_{-1} . The recurrence must begin at an index where every coefficient exists, and lower powers must be matched separately.

Statement 4. A second-order homogeneous equation normally has two independent free constants. A series built only from a_0 is one solution branch. The branch beginning with a_1 must also be included unless the equation or supplied conditions force it to vanish.

Answer 12. From Answer 4, the degree-five approximation is:

$$y(x) \approx 2 - x - 2x^2 + \frac{1}{2}x^3 + \frac{2}{3}x^4 - \frac{1}{8}x^5.$$

Set $x = 0.2$:

$$y(0.2) \approx 2 - 0.2 - 2(0.2)^2 + \frac{1}{2}(0.2)^3 + \frac{2}{3}(0.2)^4 - \frac{1}{8}(0.2)^5.$$

Calculate the powers:

$$(0.2)^2 = 0.04, \quad (0.2)^3 = 0.008, \quad (0.2)^4 = 0.0016, \quad (0.2)^5 = 0.00032.$$

Substitute them:

$$\begin{aligned} y(0.2) &\approx 2 - 0.2 - 2(0.04) + \frac{1}{2}(0.008) + \frac{2}{3}(0.0016) - \frac{1}{8}(0.00032) \\ &= 2 - 0.2 - 0.08 + 0.004 + 0.0010666667 - 0.00004 \\ &= 1.7250266667. \end{aligned}$$

Therefore:

$$\boxed{y(0.2) \approx 1.725027}.$$

This is a truncated-series approximation, not an exact decimal value.

Chapter 27 Final Summary

At an ordinary point, seek:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

For a second-order equation, $a_0 = y(x_0)$ and $a_1 = y'(x_0)$ are the two starting coefficients. Substitution, index alignment, and coefficient matching determine the rest through a recurrence.

The dependable sequence is:

check the centre → write the series → align powers
 → separate low powers → derive the recurrence
 → reconstruct the solution.